Evaluating Domain Adaptation in Financial Sentiment Analysis

FinBERT vs. Traditional NLP Baselines

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Motivation and Research Questions

Sentiment analysis is widely used in computational finance to extract market-relevant signals from financial news and corporate disclosures. However, general-domain sentiment models often perform poorly on financial language because word polarity depends strongly on domain context. Terms such as *cost*, *liability*, *debt*, or *depreciation*, for example, carry negative prior polarity in general sentiment resources [2, 6] while serving as neutral descriptive accounting terminology in financial reporting [1, 2].

This project investigates how strongly such **domain mismatch** affects 3-class financial sentiment classification (*positive*, *neutral*, *negative*) and isolates the measurable benefit of financial domain adaptation from general Transformer architecture gains. We address three research questions mapped to our experimental design:

1. **RQ1 (Domain Polarity Mismatch):** To what extent do general-purpose sentiment methods misclassify financial language, particularly by falsely assigning *negative* polarity to *neutral* descriptive statements?
2. **RQ2 (Controlled Architecture vs. Domain Adaptation):** Under a strictly leakage-free evaluation protocol, how much classification improvement stems from domain-adaptive pretraining (FinBERT) versus general-domain BERT-base and traditional machine-learning baselines?
3. **RQ3 (Linguistic Phenomenon Attribution):** Which linguistic phenomena—financial terminology, negation, hedging, numerical comparisons, or context-dependent expressions—drive model divergence?

The main contribution is not only a rigorous, leakage-free comparison of model performance, but also an interpretable phenomenon-level attribution of *when and why* domain adaptation matters in financial NLP.

Data and Experimental Setup

Accessible Datasets. All datasets and lexicons are publicly available and immediately accessible in Python. Our primary in-domain corpus is the **Financial PhraseBank** [1] (Hugging Face `financial_phrasebank`), containing 4,845 financial-news sentences annotated by 16 finance experts as *positive* ($\sim$28%), *neutral* ($\sim$59%), or *negative* ($\sim$13%). To test robustness across annotator consensus tiers, we evaluate on the 100%-agreement subset (`sentences_allagree`, 2,264 sentences), the $\geq$75%-agreement subset (`sentences_75agree`, 3,453 sentences), and the full 4,845-sentence corpus. We pair this with the **Loughran–McDonald Master Dictionary** [2] and an external out-of-distribution (OOD) holdout set, **SEntFiN 1.0** [5] (10,753 entity-sentiment annotations over financial headlines, plus FiQA headlines), filtered to unambiguous single-entity 3-class sentences and deduplicated against PhraseBank via normalized token Jaccard similarity (< 0.8).

Three-Tier Baseline Hierarchy. In Python (`PyTorch`, `transformers`, `scikit-learn`), we compare three levels:

- **Lexicon-based methods (RQ1):** General-domain **VADER** [6] valence rules versus the domain-specific **Loughran–McDonald** financial sentiment lexicon [2], with thresholds tuned strictly on validation splits.
- **Traditional machine learning (RQ1, RQ2): TF-IDF** word and n-gram features with class-weighted **Logistic Regression** and **Linear Support Vector Machines (SVM)** (1–3-grams, `sublinear_tf=True`, `fit_transform` on train folds).
- **Transformer models (RQ2, RQ3):** General-domain **BERT** (`bert-base-uncased` [4]) and finance-adapted **FinBERT** [3], sharing an identical 12-layer architecture to isolate domain pretraining.

Two-Stage Leakage-Free FinBERT Protocol. Off-the-shelf `ProsusAI/finbert` [3] was fine-tuned directly on Financial PhraseBank [1]; evaluating that checkpoint on arbitrary PhraseBank splits leaks training instances into the test set. We eliminate data leakage via a two-stage protocol: (1) **Controlled In-Domain Fine-Tuning:** On Financial PhraseBank, rather than evaluating pre-fine-tuned `ProsusAI/finbert`, we initialize from the **pre-sentiment domain-pretrained FinBERT masked language model** (pretrained only via self-supervised MLM on financial corpora prior to sentiment supervision, e.g., `yiyanghkust/finbert-pretrain` / Araci's MLM checkpoint [3]) and `bert-base-uncased` [4]. Both encoders and the TF-IDF baselines are trained, validated, and tested on the **exact same stratified 70%/15%/15% train/validation/test split** (locked via SHA-256 sentence hashes) and 5-fold stratified cross-validation under identical AdamW settings and validation Macro-F1 early stopping. (2) **External OOD Holdout Check:** To evaluate off-the-shelf `ProsusAI/finbert` alongside

our Stage 1 models without training overlap, we test all models zero-shot on deduplicated **SEntFiN 1.0** [5].

Classification Metrics and Statistical Robustness. Because this is an imbalanced 3-class classification task, we evaluate strictly with classification metrics: **Macro-F1** (primary metric), **Accuracy**, **per-class Precision, Recall, and F1-score** (focusing on false-negative confusion between the *neutral* and *negative* classes), **3×3 Confusion Matrices**, and **Expected Calibration Error (ECE)**. Pairwise differences on identical test splits are verified via paired **McNemar's test** (with continuity correction) and **paired bootstrap resampling** ($B = 10{,}000$ resamples for 95% CIs on ΔMacro-F1).

Error Analysis and Expected Contribution

Phenomenon-Stratified Error Analysis & Attribution (RQ3). Beyond aggregate metrics, we construct a **300-sentence stratified diagnostic subset** from the locked test split. Using deterministic lexical/dependency pre-tagging and double-annotation by two team members (reporting inter-annotator **Cohen's** κ), sentences are grouped into **five linguistic phenomena**: (1) *finance-specific terminology* (e.g., *liability*, *depreciation*, *amortization*), (2) *negation and scope inversion* (e.g., *did not result in a loss*), (3) *hedging and modal uncertainty* (e.g., *may*, *projected to*), (4) *numerical and directional comparisons* (e.g., *margin narrowed from 12% to 9%*), and (5) *context-dependent expressions*. For each category c, we decompose gains into the **architecture effect**, $\Delta_{\text{arch}}(c) = \text{F1}_{\text{BERT-base}}(c) - \text{F1}_{\text{TF-IDF+SVM}}(c)$, and the **domain-adaptation effect**, $\Delta_{\text{domain}}(c) = \text{F1}_{\text{FinBERT-MLM}}(c) - \text{F1}_{\text{BERT-base}}(c)$, isolating domain lexicon mastery from general capacity.

Reusable Value Produced. We will release three open-source deliverables on GitHub: (i) **leakage-free benchmark split manifests** (SHA-256 sentence hashes and PhraseBank–SEntFiN deduplication scripts), (ii) the **300-sentence phenomenon-annotated financial sentiment diagnostic dataset** with annotation guidelines, and (iii) a **reproducible Python evaluation pipeline** running all baselines, statistical tests, and error analyses.

References

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